MILITARY SPECIFICATION

COUNTER, ELECTRONIC, DIGITAL READOUT 18 GIGAHERTZ (GHZ) CONTINUOUS WAVE (CW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state automatic 18 GHz frequency counters hereinafter called equipment.

1.2 Classification. Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the following descriptions.

1.2.1 Configuration A. Configuration A equipment will be as defined in MIL-T-28800 and herein for Type II, Class 2, Style A, and forest green. Alternate colors may be specified in the contract or order.

1.2.2 Configuration B. Configuration B equipment will be as defined in MIL-T-28800 and herein for Type II, Class 4, Style C, and color optional with manufacturer. Configuration B equipment will be portable with a tilt bail handle.

1.2.3 Configuration C. Configuration C equipment will be as defined in MIL-T-28800 and herein for Type II, Class 4, Style E, and color optional with manufacturer. Configuration C equipment will be rackmountable.

1.2.4 Configuration D. Configuration D equipment will be as defined in MIL-T-28800 and herein for Type III, Class 7, Style E, and color optional with manufacturer. Except for nonoperating temperature, Configuration D equipment will comply with the Class 5 requirements of MIL-T-28800.

2. APPLICABLE DOCUMENTS.

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-C-28777	Cable Assembly, Electronic Test Equipment (3 Wire, 3 Prong, 125 Volt, Molded) Grounding Plug Connectors
MIL-T-28800	Test Equipment For Use With Electrical And Electronic Equipment, General Specification For
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, For Aerospace Applications

STANDARDS

Federal

FED-STD-595	Colors

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE-STD-1975	Standard Digital Interface for Programmable Instrumentation

(Application for copies should be addressed to the IEEE Standard Sales, 345 East 47th Street, New York NY 10017.)

3. REQUIREMENTS

3.1 Configurations and sample submissions.

3.1.1 Equipment configurations. Unless otherwise specified, requirements of this specification shall apply to all configurations. (See 6.2.1.2)

3.1.1.1 Configuration A. Configuration A equipment shall be required in MIL-T-28800 and herein for Type II, Class 2, Style A, and color forest green. Alternate colors may be specified in the contract or order. (See 3.3.3.1)

3.1.1.2 Configuration B. Configuration B equipment shall be as required in MIL-T-28800 and herein for Type II, Class 4, Style C, and Color R. Configuration B equipment shall be portable with a tilt bail handle.

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3.1.1.3 Configuration C. Configuration C equipment shall be as required in MIL-T-28800 and herein for Type II, Class 4, Style E, and Color R. Configuration C equipment shall be convertible/rackmountable.

3.1.1.4 Configuration D. Configuration D equipment shall be as required in MIL-T-28800 and herein for Type III, Class 7, Style E, and Color R. Except for non-operating temperature, Configuration D equipment shall comply with the Class 5 requirements of MIL-T-28800. (See 3.9.4.1)

3.1.2 First article or Bid Sample submission. When required in the contract or order or solicitation, sample unit(s) shall be furnished for First Article or Bid Sample inspection and approval.

3.2 Safety. The equipment shall comply with the Type III safety requirements specified in MIL-T-28800.

3.3 Parts, materials, and processes. Unless otherwise specified herein, the equipment shall comply with the Type III parts, materials, and processes specified in MIL-T-28800.

3.3.1 Restricted materials. Equipment shall comply with the restricted material requirements of MIL-T-28800 including the requirement for mercury or radioactive materials and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.3.2 Input and output signal connectors. Except for the logic interface, all input and output connectors shall be series BNC female. The logic interface connector shall be in accordance with 3.12.13.

3.3.3 Finish

3.3.3.1 Configuration A finish. Configuration A finish shall be in accordance with the Type III finish requirements of MIL-T-28800. The front panel shall be Color T and enclosure shall be forest green, Number 24052 of FED-STD-595 in accordance with MIL-C-83286. Alternate colors may be specified in contract or order. (See 6.2.1.3)

3.3.3.2 Configuration B, C, and D finish. Configurations B, C, and D finishes shall be in accordance with the Type III finish requirements of MIL-T-28800.

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements specified in MIL-T-28800.

3.4.1 Solid state construction. The equipment shall be of solid-state modular miniaturized construction. Electron tubes shall not be used.

3.4.2 Mainframe plug-in concept. Equipment externally accessible, externally removable subassemblies (mainframe plug-in concept) shall not have plug-in(s) which are removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to any such mainframe plug-in combination required to meet the requirements of this specification.

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3.4.3 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible. All potentiometer controls shall have at least a 4.7 radian (270 degrees) rotational range.

3.4.3.1 Front panel controls. Controls for the following functions shall be located on the front panel.

a. Standby - Operate

b. Sample rate

c. Resolution

d. Reset

e. Display segment test

f. Self check

j. Input port selection

3.4.4 Accessibility. The equipment shall be constructed so that:

a. Subassemblies and chassis components can be removed without removing any other hard wired printed circuit cards, subassembly, or component.

b. Adjustments can be made without removing any component, printed circuit card or subassembly except the use of extender cards is permitted.

c. Printed circuit cards can be removed without the need to unsolder cables and interconnecting wiring. Printed circuit cards (mother boards) designed primarily to distribute power and signals to other printed cards (daughter boards) are excluded from this requirement. When such mother boards are used, they shall be accessible from both sides.

3.4.5 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies shall not be used.

3.5 Electrical power sources and connections.

3.5.1 Power sources. The equipment shall comply with the Type III, 50, 60, and 400 Hertz (Hz) single phase 115/230 volt (V) alternate power sources requirements of MIL-T-28800.

3.5.2 Maximum power. The equipment shall consume not more than 150 watts.

3.5.3 Power connections. The equipment shall comply with the Type III power connections requirements of MIL-T-28800 except as specified herein.

3.5.3.1 Configuration A power cables. The power cables shall be as follows:

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a. For 115 volts alternating current (VAC) operation, the power cable shall be in accordance with MIL-C-28777/1

b. For 230 VAC operation, the power cable shall be as specified in the contract or order. (See 6.2.1.4)

3.5.3.2 Configurations B, C, and D power cables.

a. For 115 VAC operation, the power cables shall be in accordance with MIL-C-28777/3 except that the molded connector at the equipment end shall mate with Switchcraft Part Number EAC 302 connector or equal.

b. For 230 VAC operation, the power cables shall be as specified in the contract or order. (See 6.2.1.4)

3.5.4 Input power selection device. A input power selection device shall be provided for selection of input power voltages of 115 VAC or 230 VAC. Provision shall be incorporated to prevent accidental switching. The Power selection device shall be mounted on the front panel for Configuration A and on the rear panel for Configurations B, C, and D. When the equipment is delivered, the power selection device shall be in the 115 VAC position.

3.5.5 Fuses and circuit breakers. Fuses or circuit breakers shall be in accordance with MIL-T-28800. (115 VAC/230 VAC) Either common or separate fuseholders may be provided. If only one fuseholder is used (common), the equipment shall be provided with the 115 VAC fuse installed and the 230 VAC fuse shall be stowed with the accessories.

3.5.6 Input power switch. An input power switch need not be provided.

3.5.6.1 Standby-operate switch. A standby-operate switch shall be provided. The "STANDBY" position shall allow only the oven and its associated circuitry to be energized. The "OPERATE" position shall allow all circuits to be energized.

3.5.7 Power indicator. A front panel light emitting indicator(s) shall be provided to indicate when the equipment is energized. Indication shall be provided for both the standby and operate modes for Configurations A, B, and C. Indication shall be provided for the operate mode for Configuration D.

3.6 Dimensions, weight, and mechanical stability. The dimensions, weight, and mechanical stability shall be in accordance with MIL-T-28800.

3.6.1 Dimensions and weight, Configuration A

3.6.1.1 Configuration A dimensions. Maximum dimensions of Configuration A equipment with the front cover and all accessories in place shall not exceed 630 millimeters (mm) (24 inches) in width, 280m (11.0 inches) in height, and 625mm (24 inches) in depth.

3.6.1.2 Configuration A weight. The maximum weight of Configuration A equipment without the front cover and accessories shall not exceed 18 kilogram (kg) (40 pounds). The maximum weight of Configuration A equipment with the front cover and all accessories in place shall not exceed 22 kg (48 pounds).

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3.6.2 Dimensions and weight Configuration B.

3.6.2.1 Configuration B dimensions. Overall dimensions of Configuration B equipment with the front cover and all accessories inplace, excluding the tilt bail handle assembly, shall not exceed 455mm (17.9 inches) in width, 135mm (5.3 inches) in height, and 575mm (22.6 inches) in depth. The frontal area shall not exceed 455 centimeters squared. (70.7 inches squared) The tilt bail handle assembly shall not increase the width by more than 50mm (2.0 inches) and the depth by more than 75mm (3.0 inches) in the carrying position. Clearance between the front cover and any protuberances and the grip portion of the handle shall be no less than 50mm (2.0 inches).

3.6.2.2 Configuration B weight. The maximum weight of Configuration B equipment without the front cover and accessories shall not exceed 16kg (35 pounds). The maximum weight of Configuration B equipment with the front cover and all accessories in place shall not exceed 18kg (40 pounds)

3.6.3 Weight and dimensions Configuration C.

3.6.3.1 Configuration C weight. The maximum weight of Configuration C equipment shall not exceed 16kg (25 pounds).

3.6.3.2 Configuration C dimensions. Partial dimensions of Configuration C equipment behind the front panel shall not exceed 455mm (17.9 inches) in depth, 134mm (5.25 inches) in height, and 447.6mm (17.625 inches) in width including slides. Nominal size of the front panel when converted for rackmounting shall be 134mm (5.25 inches) high and in accordance with the rackmounting requirements of MIL-T-28800.

3.6.4 Weight and dimensions Configuration D.

3.6.4.1 Configuration D weight. The maximum weight of Configuration D equipment shall not exceed 14kg (20.0 pounds).

3.6.4.2 Configuration D dimensions. Maximum dimensions of Configuration D equipment shall not exceed 483mm (19 inches) wide by 140mm (5.5 inches) high by 500mm (19.7 inches) in depth. Frontal area shall not exceed 500 centimeters squared (80 inches squared).

3.7 Enclosures. Configurations A, B, and C enclosures shall be designed to facilitate horizontal viewing for bench use and vertical viewing when the equipment is placed on the floor.

3.7.1 Configuration A enclosures. The Configuration A enclosure shall be as specified in MIL-T-28800 for Style A.

3.7.2 Configuration B enclosures. The Configuration B enclosure shall be as specified in MIL-T-28800 for Style C and shall be supplied with a removable tilt bail handle in lieu of Style C handle specified in MIL-T-28800. The handle shall permit the equipment to be carried face up with the longest dimension vertical and shall tilt and support the chassis for easier viewing. The design of the handle requires procuring activity approval (see 6.2.2.13).

3.7.2.1 Configuration B stacking. The Configuration B enclosure shall comply with the stacking provisions specified for Style C in MIL-T-28800 for a minimum of three equipments. The tilt bail handle shall not impede the simultaneous operation, viewing, and stacking of the equipment.

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3.7.3 Configuration C enclosure. The Configuration C enclosure shall be as specified in MIL-T-28800 for Style E and shall be convertible/rackmountable. The equipment shall be configured for rackmounting and shall include all necessary hardware including any filler panels required to conform to 3.6.3.2.

3.7.3.1 Rackmount slides. Rackmount slides shall be supplied (but not necessarily installed) with Configuration C equipment to permit access to the rear panel of the equipment when rackmounted and in the extended position (See 6.2.1.5).

3.8 Marking and identification. The equipment shall comply with the Type III marking and identification requirements of MIL-T-28800. (See 6.2.2.1 and 6.2.2.2)

3.8.1 System Marking. The system part number shall be marked on the major assembly identification plate and on any externally removable subassemblies furnished in accordance with 3.4.2.

3.9 Environmental requirements. Equipment shall comply with the environmental requirements of MIL-T-28800 as specified herein. (See 6.2.2.14)

3.9.1 Configuration A environmental requirements. Configuration A equipment shall comply with the Class 2, Style A environmental requirements of MIL-T-28800 and as specified herein.

3.9.1.1 Configuration A salt atmosphere. Configuration A equipment shall comply with the Style A salt atmosphere exposure requirements of MIL-T-28800 for enclosure and structural parts.

3.9.2 Configuration B environmental requirements. Configuration B equipment shall comply with the Class 4, Style C requirements of MIL-T-28800 and as specified herein.

3.9.2.1 Configuration B salt atmosphere. Configuration B equipment shall comply with the Style C salt atmosphere exposure requirements of MIL-T-28800 for structural parts.

3.9.3 Configuration C environmental requirements. Configuration C equipment shall comply with the Class 4, Style E requirements of MIL-T-28800 and as specified herein.

3.9.3.1 Configuration C Salt atmosphere. Configuration C equipment shall comply with the Style E salt atmosphere exposure requirements of MIL-T-28800 for structural parts.

3.9.4 Configuration D environmental requirements. Except for nonoperating storage temperature, Configuration D equipment shall comply with the Class 5, Style E requirements of MIL-T-28800.

3.9.4.1 Configuration D nonoperating storage temperature. Configuration D equipment shall comply with the Class 5 nonoperating storage temperature requirements of MIL-T-28800 except the lower temperature limit shall be -40° Celcius (C).

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3.9.5 Electromagnetic interference (EMI).

3.9.5.1 EMI. Unless otherwise specified in the contract or order, the equipment shall comply with the EMI requirements of MIL-T-28800 and as specified in a thru i.

a. CE 01: Narrow band requirements from 10 kilohertz (kHz) to 20 kHz only

b. CE 03: Full limits.

c. CS 01: Full limits.

d. CS 02: Full limits.

e. CS 06: Full limits.

f. RE 01: Relaxed 10 decibel (dB) at the fundamental, third harmonic, and fifth harmonic of the power source frequency.

g. RE 02: Full limits to 10 times the highest used or intentionally generated frequency or 1 Gigahertz (GHz), whichever is greater.

h. RS 01: Full limits.

i. RS 03: 1 volt/meter limit to 1 GHz.

3.9.6 Magnetic environment. All configurations shall comply with the DC magnetic environment requirements of MIL-T-28800. There shall be no degradation in fields up to 5 Oersteds and damage to the equipment shall not occur in the presence of up to a 20 Oersted field. Normal operation of the equipment shall automatically resume when the externally induced magnetic field is removed.

3.9.7 Warm-up. Warm-up shall be as specified in 3.12.2.

3.10 Reliability. Reliability shall comply with the Type III requirements of MIL-T-28800 and as specified herein. The specified mean time between failures (MTBF) shall be 5000 hours.

3.10.1 Reliability requirements. Each equipment delivered against this specification shall be subjected to a burn in of not less than 100 hours at 40° C of which at least the last 30 hours shall be failure free (See 6.2.3.1).

3.11 Maintainability requirements. The equipment shall comply with the Type III maintainability requirements of MIL-T-28800 and as specified herein (See 6.2.3.2).

3.11.1 Certification time. Certification time shall not exceed 2.0 hours, and it shall include all time required to check all inputs and outputs, on all range function settings.

3.11.2 Maintenance provisions. The equipment shall meet the maintenance provisions of MIL-T-28800 except the accessibility of accessories does not apply.

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3.11.2.1 Maintenance and calibration aids. When required in the contract or order, any circuit board extenders, special adapters, special tools, and patch cables required for maintenance or calibration of equipment shall be stowed in a suitable reusable pouch or container which shall be marked "MAINTENANCE AIDS FOR USE WITH (equipment nomenclature or model number)" (See 6.2.1.7). The equipment nomenclature shall be used thereon if assigned (See 6.2.2.1).

3.11.3 Washability. The equipment shall comply with the washability requirements of MIL-T-28800 (See 6.2.2.9).

3.11.4 Tilt bail handle, Configuration B. The tilt bail handle assembly shall permit compliance with the enclosure attitude maintenance provisions of MIL-T-28800. Removal of the handle or handle assembly with common tools is permitted to comply with this requirement.

3.12 Performance requirements. The equipment shall contain the following electrical performance capabilities.

a.	Environmental performance	(3.12.1)
b.	Warm-up performance	(3.12.2)
c.	Time base	(3.12.3)
d.	Display	(3.12.4)
e.	Sample rate	(3.12.5)
f.	Self check	(3.12.6)
g.	Frequency measurement	(3.12.8)
h.	Input characteristics	(3.12.7)
i.	Outputs	(3.12.9)
j.	Logic interface	(3.12.10)
k.	Accessories	(3.12.11)

3.12.1 Environmental performance. Equipments shall meet the requirements specified herein under all combinations of environments specified herein.

3.12.2 Warm-up performance. Warm-up time shall not exceed one minute after the operate mode is selected when sufficient time has previously elapsed in the standby mode to permit the time base to stabilize to the degree required for the measurement in accordance with 3.12.3.

3.12.3 Time base. Time base specifications are given in terms of fractional frequency error (ffe) (See 6.3.1). Except for temperature stability, all time base requirements are specified for 25°C. Time base requirements are specified for measurement from the 10 megahertz (MHz) standard output with the equipment in the self test mode (See 4.5).

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3.12.3.1 Time base high stability. Unless otherwise required in the contract or order, the equipment shall comply with the following requirements.

3.12.3.1.1 Long term stability (aging rate), high stability. The aging rate shall be less than $\pm 5 \times 10^{-10}$ per 24 hours after 48 hours of operation (M_8, M_9) preceded by at least 72 hours with no source power applied.

3.12.3.1.2 Frequency retrace, high stability. The ffe 30 minutes after application of source power (M_2), following at least a 72 hour period of nonoperation, shall be within $\pm 2 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The ffe 5 hours after application of source power (M_4), following at least a 72 hour period of nonoperation, shall be within $\pm 1 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The 72 hour period of nonoperation shall be preceded by at least 72 hours of operation.

3.12.3.1.3 Restabilization, high stability. The ffe 5 hours after application of source power (M_6) shall be within $\pm 5 \times 10^{-9}$ of the ffe measured 48 hours after application of source power (M_7). Application of source power shall be preceded by at least 72 hours of nonoperation.

3.12.3.1.4 Short term stability, high stability. The short term stability shall be less than $\pm 5 \times 10^{-10}$ root mean square (rms) 30 minutes after application of source power (M_3) and less than $\pm 5 \times 10^{-11}$ rms 5 hours after application of source power (M_5). Application of source power shall be preceded by at least 72 hours with no source power applied.

3.12.3.1.5 Stability during power change, high stability. The ffe shall not change more than $\pm 1 \times 10^{-8}$ during the first two minutes after any power change within the specified steady state band nor more than 1×10^{-10} at the end of the two minutes (See 3.5.1).

3.12.3.1.6 Stability during temperature change, high stability. The ffe shall not change more than $\pm 7 \times 10^{-9}$ total for any incremental or total temperature change within the specified temperature operating range (See 3.9).

3.12.3.1.7 Total ffe, high stability. The total calculated ffe at the end of a 12 month period shall not exceed $\pm 2 \times 10^{-7}$

3.12.3.1.8 Oscillator adjustments, high stability. The oscillator shall be capable of being adjusted so that the 10 MHz standard output shall be within ±1 part in 10^9 of an external 10 MHz standard. The total range of adjustment shall be at least ±1 part in 10^6 from 10 MHz.

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3.12.3.2 Time base, low stability. When required in the contract or order, the equipment shall comply with the following characteristics (See 6.2.1.8).

3.12.3.2.1 Long term stability (aging rate), low stability. The aging rate shall be less than $\pm 1 \times 10^{-9}$ per 24 hours after 48 hours of operation (M_8, M_9) preceded by at least 72 hours with no source power applied.

3.12.3.2.2 Frequency retrace, low stability. The ffe 30 minutes after application of source power (M_2), following at least a 72 hour period of nonoperation, shall be within $\pm 4 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The ffe 5 hours after application of source power (M_4), following at least a 72 hour period of nonoperation, shall be within $\pm 2 \times 10^{-8}$ of the ffe immediately prior to the 72 hour period of nonoperation (M_1). The 72 hour period of nonoperation shall be preceded by at least 72 hours of operation.

3.12.3.2.3 Restabilization, low stability. The ffe 5 hours after application of source power (M_6) shall be within $\pm 1 \times 10^{-8}$ of the ffe measured 48 hours after application of source power (M_7). Application of source power shall be preceded by at least 72 hours of nonoperation.

3.12.3.2.4 Short term stability, low stability. The short term stability shall be less than 1×10^{-9} rms 30 minutes after application of source power (M_3) and less than 1×10^{-10} rms 5 hours after application of source power (M_5). Application of source power shall be preceded by at least 72 hours with no source power applied.

3.12.3.2.5 Stability during power change, low stability. The ffe shall not change more than $\pm 2 \times 10^{-8}$ during the first two minutes after any power change within the specified steady state band nor more than $\pm 2 \times 10^{-10}$ at the end of the two minutes (See 3.5.1).

3.12.3.2.6 Stability during temperature change, low stability. The ffe shall not change more than $\pm 1.4 \times 10^{-8}$ total for any incremental or total temperature change within the specified temperature operating range (See 3.9).

3.12.3.2.7 Total ffe, low stability. The total calculated ffe at the end of a 12 month period shall not exceed $\pm 4 \times 10^{-7}$.

3.12.3.2.8 Oscillator adjustment, low stability. The oscillator shall be capable of being adjusted so that the 10 MHz standard output shall be within ±1 part in 10^9 of an external 10 MHz standard. The total range of adjustment shall be at least ±1.5 parts in 10^6 from 10 MHz.

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3.12.3.3 External standard input. Configuration A, B, and C equipment shall be capable of specified operation from any operator selected external frequency standard of 1 MHz, 5 MHz, or 10 MHz having a value of 1 volt rms into 50 ohms. Configuration D equipment shall be capable of specified operation from any 10 MHz external frequency standard having a value of 1 volt rms into 50 ohms. When the external standard is used, it shall be in lieu of the internal standard.

3.12.3.3.1 External standard input impedance. The load impedance at the equipment external standard input port shall be not less than 1 kilohm paralleled by not more than 30 picofarad (pf).

3.12.4 Display. All front panel visible light or glow devices are display devices and shall be readable at a distance of 15 feet through a subtended angle of 45 degrees in an ambient light level of 100 foot candles. Provisions shall be incorporated to test all light emitting segments of the display devices.

3.12.4.1 Numeric readout. The numeric readout shall be a 9 character, in line, horizontal, single plane, digital display at least 10 mm (0.4 inches) high.

3.12.4.1.1 Measurement units. Measurement units, in accordance with a through d, shall be presented in conjunction with the numeric display.

a. GHz

b. Mhz

c. KHz

d. Hz

3.12.4.1.2 Display storage. The numeric readout shall be continuous and change only when the measurement is different from that counted during the previous measurement cycle.

3.12.4.1.3 Display blanking. All non-significant leading zeros shall be blanked. All readout digits below the selected resolution of the displayed frequency shall not be displayed.

3.12.4.2 Gate lamp. A front panel display device shall indicate gate opening and closing.

3.12.4.3 Overload indicator. A flashing overload display device for Configurations A, B, and C shall warn of signal levels approaching the maximum operating input.

3.12.5 Sample rate. Except when the logic interface is required, the sample rate shall be variable from 0.1 to at least 5 seconds. When the logic interface is required, the sample rate shall be variable from 0.01 to at least 5 seconds (See 3.12.13). The sample rate shall be capable of being held indefinitely and shall be independent of the resolution control. Erroneous data resulting from conflicts between sample rate and gate time shall not be displayed (See 6.3.3).

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3.12.5.1 Reset. The reset control shall allow manual initiation of the rate.

3.12.6 Self check. The self check function shall be provided to verify functional operation of the frequency measurement mode.

3.12.7 Frequency measurement.

3.12.7.1 Frequency range. The frequency range shall be at least 20 Hz to 18 GHz.

3.12.7.2 Frequency accuracy. The frequency measurement accuracy shall be ± one count ± time base stability.

3.12.7.3 Frequency resolution. Resolution from 1 Hz to 1 MHz in decade multiples shall be provided.

3.12.7.4 F. M. Tolerance. The equipment shall tolerate and meet the accuracy requirements of 3.12.7.2 for signals between 500 MHz and 18 GHz which have peak to peak deviation of up to 10 MHz at a deviation rate from zero to at least 10 MHz.

3.12.7.5 A. M. Tolerance. The equipment shall tolerate and correctly measure an input signal which is sine wave amplitude modulated with frequencies up to 1 KHz and has a modulation index not greater than 50 percent.

3.12.7.6 Signal discrimination. Discrimination between desired and undesired signals shall be provided in accordance with a or b.

a. The equipment shall lock onto and measure the signal which is 20 dB greater in amplitude than any other signal.

b. The equipment shall lock onto and measure the lowest frequency signal above the operator selected start frequency (comb line).

3.12.7.7 Measurement time. At one hertz resolution the equipment shall lock onto the measure the input signal in not more than 1.5 seconds.

3.12.8 Input characteristics. A maximum of three input ports shall be provided with a switching device for operator selection of the desired input port.

3.12.8.1 Input impedance. The high impedance input port(s) shall be at least from 20 Hz to 20 MHz and shall be at least 1 megohm shunted by not more than 35 pf. The low impedance input port(s) shall be at least from the upper frequency of the high impedance input port with at least a 10 percent overlap to 18 GHz. The impedance of the low impedance input port(s) shall be 50 ohms nominal.

3.12.8.2 Voltage standing wave ratio (VSWR). The VSWR of the 50 ohm input port (s) shall not exceed 2:1 up to 12.4 GHz and 3:1 from 12.4 GHz to 18 GHz.

3.12.8.3 Coupling. AC coupling shall be provided for all input ports.

3.12.8.4 Sensitivity. The equipment sensitivity of the high impedance input port(s) shall be at least 50 millivolts (mV) root mean square (RMS). The equipment sensitivity of the low impedance input port(s) shall be at least -20 dBm up to 850 MHz and at least -25 dBm from 250 MHz to 18 GHz.

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3.12.8.5 Maximum operating input. The equipment shall operate satisfactorily with input signals of up to 120 volts RMS decreasing at a rate of 6 dB per octave down to 2 volts RMS on the high impedance input port(s) and up to +7 dBm on the low impedance input port(s).

3.12.8.6 Maximum input without damage. The equipment shall accept without damage input signals of up to 150 volts RMS decreasing at a rate of 6 dB per octave down to 2 volts RMS on the high impedance input port(s) and up to +30 dBm on the low impedance port(s).

3.12.9 Outputs.

3.12.9.1 Standard frequency outputs, Configuration A, B, C. Configurations A, B, and C equipment shall provide (in both the standby and operate mode) 1 MHz and 10 MHz standard frequency outputs from the controlling standard which are independent of the time base control setting and are uninterrupted during all functions. The 1 MHz and the 10 MHz standard outputs need not be available simultaneously. The output voltage shall be at least 1 volt RMS into an external impedance of 50 ohms for each frequency.

3.12.9.2 Standard frequency output, Configuration D. Configuration D equipment shall provide a 1 MHz standard frequency output which is independent of the time base switch setting and uninterrupted during all functions.

3.12.10 Logic interface. When required in the contract or order, a logic interface shall be provided (See 6.2.1.10). When a logic interface is required, it shall be in accordance with IEEE-STD-488-1975.

3.12.10.1 Function programming characteristics. All measurement data, including multipliers and units, shall be transferred by means of the logic interface. All control and switch functions except for the following shall be programmable.

a. Source power selection.

b. Frequency standard selection.

c. Standby/operate selection.

3.12.10.1.2 Sample programming. Under interface control the equipment shall have two sampling modes (as selected by the controller). In the first mode the equipment shall not make any measurement until commanded to do so by the controller. The rate at which the controller can command the equipment to make measurements shall be at least 100 measurements per second or the maximum rate possible with the required resolution.

In the second mode the equipment shall continually make measurements on its own at a rate of at least 100 measurements per second or at the maximum rate possible with the required resolution.

3.12.10.2 Interface characteristics. The equipment shall include the interface functions as in a thru f.

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a. Source handshake interface function (SH), subset identification - SH1.

b. Acceptor handshake interface function (AH), subset identification - AH 1.

c. Talker interface function (T), subset identification - T1.

d. Listener interface function (L), subset identification - L2.

e. Service request interface function (SR), subset identification - SR1.

f. Remote local interface function (RL), subset identification - RL1.

3.12.10.2.1 Numeric data. Unless otherwise specified in the contract or order, all numeric data (input and output) shall be American Standard Code for Information Interchange (ASCII) and the most significant digit shall be transferred first.

3.12.10.2.2 Alpha-numeric data. Unless otherwise specified in the contract or order, all alpha-numeric data (input and output) shall be ASCII.

3.12.10.2.3 Address assignment. Complete five bit switchable address capabilities shall be provided for the talker function (TI) address (MTA) and the listener function (L2) address (MLA) on the front panel for Configuration A and on the rear panel for Configurations B, C, and D. If a single device for address switching is provided for both the MLA and the MTA, then the five bit switchable address for the MLA shall be equal to that for the MTA.

3.12.11 Accessories. The following accessories shall be supplied with each equipment.

a. Power cable in accordance with 3.5.3.1a for Configuration A and 3.5.3.2a for Configurations B, C, and D.

b. One operator's manual.

c. Fuse in accordance with 3.5.5.

3.12.14.1 Stowage. Accessories and maintenance and calibration aids (if required, see 3.11.2.1) shall be stowed in the front cover for Configurations A and B and in an external detachable pouch for Configurations C and D.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Bid Sample testing will be performed by the Government.

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4.1.1 Quality assurance terms and definitions. Qualify assurance terms used herein shall be as in MIL-T-28800.

4.2 Classification of inspections. The inspections specified herein are classified as in (a) through (c).

(a) First Article inspection (see 3.1.2 and 6.2.1).

(b) Bid Sample inspection (see 3.1.2 and 6.2.1).

(c) Quality conformance inspection (see 4.2.3).

4.2.1 First Article inspection groups. First Article examinations and tests are classified into groups in accordance with Table I, 4.3, and (a) through (f) (see 6.2.1.11).

(a) First Article inspection, Group A (see 4.3.1.1).

(b) First Article tests, Group B (see 4.3.1.2).

(c) First Article tests, Group C (see 4.3.1.3).

(d) First Article tests, Group D (see 4.3.1.4).

(e) First Article tests, Group E (see 4.3.1.5).

(f) First Article tests, Group F (see 4.3.1.6).

4.2.2 Bid Sample inspection groups. Bid Sample examinations and tests are classified into groups in accordance with Table I, 4.3, and (a) through (f) (see 6.2.1.12).

(a) Bid Sample inspection, Group A (see 4.3.2.1).

(b) Bid Sample tests, Group B (see 4.3.2.2).

(c) Bid Sample tests, Group C (see 4.3.2.3).

(d) Bid Sample tests, Group D (see 4.3.2.4).

(e) Bid Sample tests, Group E (see 4.3.2.5).

(f) Bid Sample tests, Group F (see 4.3.2.6).

4.2.3 Quality conformance inspection groups. Quality conformance examinations and tests are classified into groups in accordance with Table I, 4.4, and (a) through (f).

(a) Production inspection, Group A (see 4.4.1).

(b) Production control tests, Group B (see 4.4.2).

(c) Environmental and electrical power tests, Group C (see 4.4.3).

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TABLE I Examination and test groups

Description	Requirement	Inspection Method: First Article	Bid Sample	Quality Conformance
Group A				
Preoperational inspection	3	4.5	4.5	4.5
Level A performance	3.12	4.5	4.5	4.5
Leakage current	3.2	4.5	4.5	4.5
Group B				
Level B performance	3.12	4.5	4.5	4.5
Group C				
Temperature and humidity	3.9	4.5	4.5	4.5
Humidity	3.9	4.5	4.5	4.5
Low temperature	3.9	4.5	4.5	4.5
High temperature	3.12.3.6	4.5	4.5	4.5
Temperature stability, Time Base	3.9	4.5	4.5	4.5
Altitude	3.9	4.5	4.5	4.5
Vibration				
Shock	3.9	4.5	4.5	4.5
Shock pulse				
Transit drop, Configurations A & B	3.9	4.5	4.5	4.5
Bench handling	3.9	4.5	4.5	4.5
Water resistance				
Watertight, Configuration A	3.9	4.5	4.5	4.5
Splashproof, Configuration A	3.9	4.5	4.5	4.5
Dripproof, Configuration B	3.9	4.5	4.5	4.5
Electrical power				
Input power consumption	3.5.2	4.5	4.5	4.5
Voltage and frequency variation	3.5.1	4.5	4.5	4.5
Voltage transient	3.5.1	4.5	4.5	4.5
Frequency transient	3.5.1	4.5	4.5	4.5
Line voltage stability, Time Base	3.12.3.5	4.5	4.5	4.5
Verification, Group C	3.12	4.5	4.5	4.5

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TABLE I - Examination and test groups. - Continued

Description	Requirement	Inspection Method		
		First Article	Bid Sample	Quality Conformance
Group D				
Fungus resistance (material)	3.9	4.5	4.5	4.5
Salt atmosphere exposure				
Structural parts				
Configuration A	3.9.1.1	4.5	4.5	4.5
Configuration B	3.9.2.1	4.5	4.5	4.5
Configuration C	3.9.3.1	4.5	4.5	4.5
Enclosures				
Configuration A	3.9.1.1	4.5	4.5	4.5
Explosive atmosphere				
Configuration A	3.9.1	4.5	4.5	4.5
Sand and dust				
Configuration A	3.9.1	4.5	4.5	4.5
EMI				
Configuration A, B, & C	3.9.5.1	4.5	4.5	4.5
Configuration D	3.9.5.2	4.5	4.5	4.5
Magnetic Environment	3.9.6	4.5	4.5	4.5
Verification, Group D				
Configuration A	3.12	4.5	4.5	4.5
Group E				
Maintainability	3.11	4.5	4.5*	4.5
Washability	3.11.4	4.5	4.5*	4.5
Acoustic noise	3.9	4.5	4.5	4.5
Dimensions	3.6	4.5	4.5	4.5
Weight	3.6	4.5	4.5	4.5
Mechanical stability	3.6	4.5	4.5	4.5
Front panel marking	3.8	4.5	4.5	4.5
Group F				
Reliability	3.10	4.5	4.5*	4.5

* See 4.3.2.5a and 4.3.2.6

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(d) Environmental tests, Group D (see 4.4.4).

(e) Other tests, Group E (see 4.4.5).

(f) Reliability tests, Group F (see 4.4.6).

4.3 First Article or Bid Sample inspection. Unless otherwise required herein, the test methods, procedures and conditions used for First Article or Bid Sample inspection shall be performed on the units as specified in 4.5 and MIL-T-28800 (see 6.2.2.3, 6.2.2.4, and 6.2.2.5).

4.3.1 First Article inspection. When required in the solicitation, the First Article inspection shall be performed on five equipment and on a sample of the front panel (see 6.3.9).

4.3.1.1 First Article inspection Group A. The five equipment shall be subjected to the inspection specified in Table I and II.

4.3.1.2 First Article tests, Group B. The five equipment that have satisfactorily passed the required tests in accordance with 4.3.1.1 shall be subjected to the Group B tests of Tables I and III.

4.3.1.3 First Article tests, Group C. Three equipment that have satisfactorily passed the Group B tests shall be subjected to the Group C tests of Table I.

4.3.1.4 First Article tests, Group D. Two equipment that have satisfactorily passed the Group B tests shall be subjected to the Group D tests of Table I.

4.3.1.5 First Article tests, Group E. Equipment that have satisfactorily passed the Group B tests shall be subjected to the Group E tests of Table I. The number of equipment tested shall be as in (a) through (e).

(a) Maintainability. One equipment shall be tested. The washability test shall be performed on the one equipment prior to the maintainability demonstration.

(b) Acoustic noise. One equipment shall be tested, except that equipment that do not contain a motor (such a fan or blower) need not be tested.

(c) Dimensions and weight. One equipment shall be tested.

(d) Mechanical stability. One equipment shall be tested.

(e) Front panel marking. One 100mm (4 inch) by 100mm (4 inch) sample, which is representative of the equipment front panel and marking, shall be tested.

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TABLE II LEVEL A PERFORMANCE PARAMETERS*

Parameter	Requirement
Display	3.12.4
Sample Rate	3.12.5
Self Check	3.12.6
Frequency Measurement	3.12.7
Input Characteristics	3.12.8
Output	3.12.10
Logic Interface (when required)	3.12.11

*Reduced testing

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TABLE III LEVEL B PERFORMANCE PARAMETERS

Parameter	Requirement
Warm-up	3.12.2
Time base	3.12.3
Time base, high stability	3.12.3.1
Long term stability	3.12.3.1.1
Frequency retrace	3.12.3.1.2
Restabilization	3.12.3.1.3
Short term stability	3.12.3.1.4
Stability during power change	3.12.3.1.5
Stability during temperature change	3.12.3.1.6
Total ffe	3.12.3.1.7
Oscillator adjustment	3.12.3.1.8
Time base, low stability (when required)	3.12.3.2
Long term stability	3.12.3.2.1
Frequency retrace	3.12.3.2.2
Restabilization	3.12.3.2.3
Short term stability	3.12.3.2.4
Stability during power change	3.12.3.2.5
Stability during temperature change	3.12.3.2.6
Total ffe	3.12.3.2.7
Oscillator adjustment	3.12.3.2.8
External standard input	3.12.3.3
External standard input impedance	3.12.3.3.1
Display	3.12.4
Numeric readout	3.12.4.1
Measurement units	3.12.4.1.1
Display storage	3.12.4.1.2
Display blanking	3.12.4.1.3
Gate lamp	3.12.4.2
Overload indicator	3.12.4.3
Sample rate	3.12.5
Reset	3.12.5.1
Self check	3.12.6
Frequency measurement	3.12.7
Frequency range	3.12.7.1
Frequency accuracy	3.12.7.2
Frequency resolution	3.12.7.3
F.M. tolerance	3.12.7.4
A.M. tolerance	3.12.7.5
Signal discrimination	3.12.7.6
Measurement time	3.12.7.7

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TABLE III LEVEL B PERFORMANCE PARAMETERS (CONT'd)

Parameter	Requirement
Input characteristics	3.12.8
Input impedance	3.12.8.1
VSWR	3.12.8.2
Coupling	3.12.8.3
Sensitivity	3.12.8.4
Maximum operating input	3.12.8.5
Maximum input without damage	3.12.8.6
Outputs	3.12.9
Standards outputs, Configurations A, B, and C	3.12.9.1
Standard output, Configuration D	3.12.9.2
Gate output, Configurations A, B, and C	3.12.9.3
Gate output, Configurations D	3.12.9.4
Logic interface	3.12.10
Function programming characteristics	3.12.10.1
Sample programming	3.12.10.1.1
Interface characteristics	3.12.10.2
Numeric data	3.12.10.2.1
Alph - numeric data	3.12.10.2.2
Address assignment	3.12.10.2.3

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4.3.1.6 First Article tests, Group F. The Group F reliability demonstration test shall be conducted on all 5 equipment in accordance with the Type III demonstration phase of MIL-T-28800.

4.3.2 Bid Sample inspection. Procurement of equipment by the Bid Sample method shall require the submission of a Bid Sample to the procuring activity when required in the solicitation. Unless otherwise required in the solicitation, the Bid Sample shall consist of two equipment with accessories and instruction manuals, a front panel sample (see 6.3.9), and clear and concise rationale showing how the reliability and maintainability characteristics of the equipment comply with the requirements of the solicitation. The Bid Sample shall be representative of equipment specified by the requirements of the solicitation (see 6.2.1.14).

4.3.2.1 Bid Sample inspection, Group A. The two Bid Sample equipment will be subjected to the inspection specified in Table I and II.

4.3.2.2 Bid Sample tests, Group B. The two Bid Sample equipment that have satisfactorily passed the required tests in accordance with 4.3.2.1 will be subjected to the Group B tests of Tables I and III.

4.3.2.3 Bid Sample tests, Group C. One Bid Sample equipment that has satisfactorily passed the required tests in accordance with 4.3.2.2 will be subjected to the Group C tests of Table I.

4.3.2.4 Bid Sample tests, Group D. One Bid Sample equipment that has satisfactorily passed the required tests in accordance with 4.3.2.2 will be subjected to the Group D tests of Table I.

4.3.2.5 Bid Sample tests, Group E. Equipment that have satisfactorily passed the required tests in accordance with 4.3.2.2 will be subjected to the Group E tests of Table I. The number of equipment tested will be as in (a) through (e).

(a) Maintainability. Unless otherwise specified in the solicitation, no equipment will be tested.

(b) Acoustic noise. One equipment will be tested, except that equipment that do not contain a motor (such as a fan or blower) will not be tested.

(c) Dimensions and weight. One equipment will be tested.

(d) Mechanical stability. One equipment will be tested.

(e) Front panel marking. One sample of the front panel will be tested.

4.3.2.6 Bid Sample tests, Group F. Unless otherwise specified in the solicitation, Group F reliability testing will not be performed on Bid Samples.

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4.4 Quality conformance inspection. Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as specified in 4.5 and MIL-T-28800. (See 6.2.2.11 and 6.2.2.12).

4.4.1 Production inspection. Each equipment shall be subjected to the Group A inspection of Tables I and II to ensure qualitatively the proper functioning of the equipment, including all operating controls, and conformance with safety requirements.

4.4.2 Production control tests. The Group B tests of Tables I and III shall be performed on equipment that have passed Group A inspection. (See 4.4.1)

4.4.2.1 Production control sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group B tests. Minimum sampling shall be in accordance with MIL-T-28800.

4.4.3 Environmental and electrical power tests. The Group C tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.3.1 Environmental and electrical power sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group C tests. The sampling shall be in accordance with MIL-T-28800.

4.4.4 Environmental tests. The Group D tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.4.1 Environmental tests sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group D tests. The sampling shall be in accordance with MIL-T-28800.

4.4.5 Other tests. The Group E tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.5.1 Other tests sampling plan. From the first production lot, a random sample shall be selected and subjected to the Group E tests. The sampling shall be in accordance with (a) through (d).

(a) Maintainability. One equipment.

(b) Acoustic noise. One equipment.

(c) Dimensions and weight. One equipment.

(d) Mechanical stability. One equipment.

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4.4.6 Reliability tests. The Group F tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2. The Group F tests shall be conducted in accordance with the Type III reliability tests of MIL-T-28800. Burn in for each reliability sample unit shall be a minimum of 100 hours with the last 30 hours failure free. If a failure occurs within the 30 hour failure free period, burn in on that unit will stop, repair will be made, and an additional 30 hours burn in will be added to the total accrued testing time for that unit. Additional failures shall result in additional burn in extensions until the final 30 hours of burn in are failure free.

4.4.6.1 Reliability demonstration phase. Unless reliability has been demonstrated in accordance with 4.3.1.6, the demonstration phase of the Type III reliability test shall be performed on a random sample of ten equipment selected from the first production lot. If reliability has been demonstrated in accordance with 4.3.1.6, the first production lot need not undergo the demonstration phase of the Type III reliability test.

4.4.6.2 Reliability sampling phase. If the reliability has been demonstrated in accordance with 4.3.1.6, the sample phase of the Type III reliability tests shall be performed on a random sample of 10 equipment selected from each production lot. If reliability has been demonstrated on the first production lot in accordance with 4.4.6.1, subsequent production lots shall undergo the sampling phase of the Type III reliability test.

4.5 Examination and test methods. Examination and test methods shall be in accordance with MIL-T-28800 and as specified herein (see 6.2.2.6, 6.2.2.7, 6.2.2.8, 6.2.2.10, 6.2.3.3, Tables I, II, and III). Configurations A, B, and C shall be tested for the applicable class requirements of MIL-T-28800. Except for nonoperating storage range, Configuration D shall be tested for the Class 5 requirements of MIL-T-28800 (see 3.9.4.1). When the equipment is identified by Configuration, it shall be for the applicable Type, Class, and Style as required in 3.1.1 through 3.1.4. The time base testing shall be in accordance with Figure 1. Equipment operation shall be verified in accordance with a and b below:

a. Satisfactory operation check. The satisfactory operation check shall confirm the proper operation of the internal time base, the input circuits of at least one channel, and the display section of the equipment.

b. Satisfactory operation test. The satisfactory operation test shall confirm the specified operation of the equipment for all parameters of the Level A performance test (see Table II).

4.5.1 Magnetic environment test. The magnetic environment test shall be performed in accordance with MIL-T-28800. If the magnitude of the 20 Oersted field is reduced as permitted in the test, the reduction may be to a 5 Oersted level. Degradation of the equipment performance in a 5 Oersted field or permanent damage to the equipment resulting from the 20 Oersted field, shall constitute a failure of this test.

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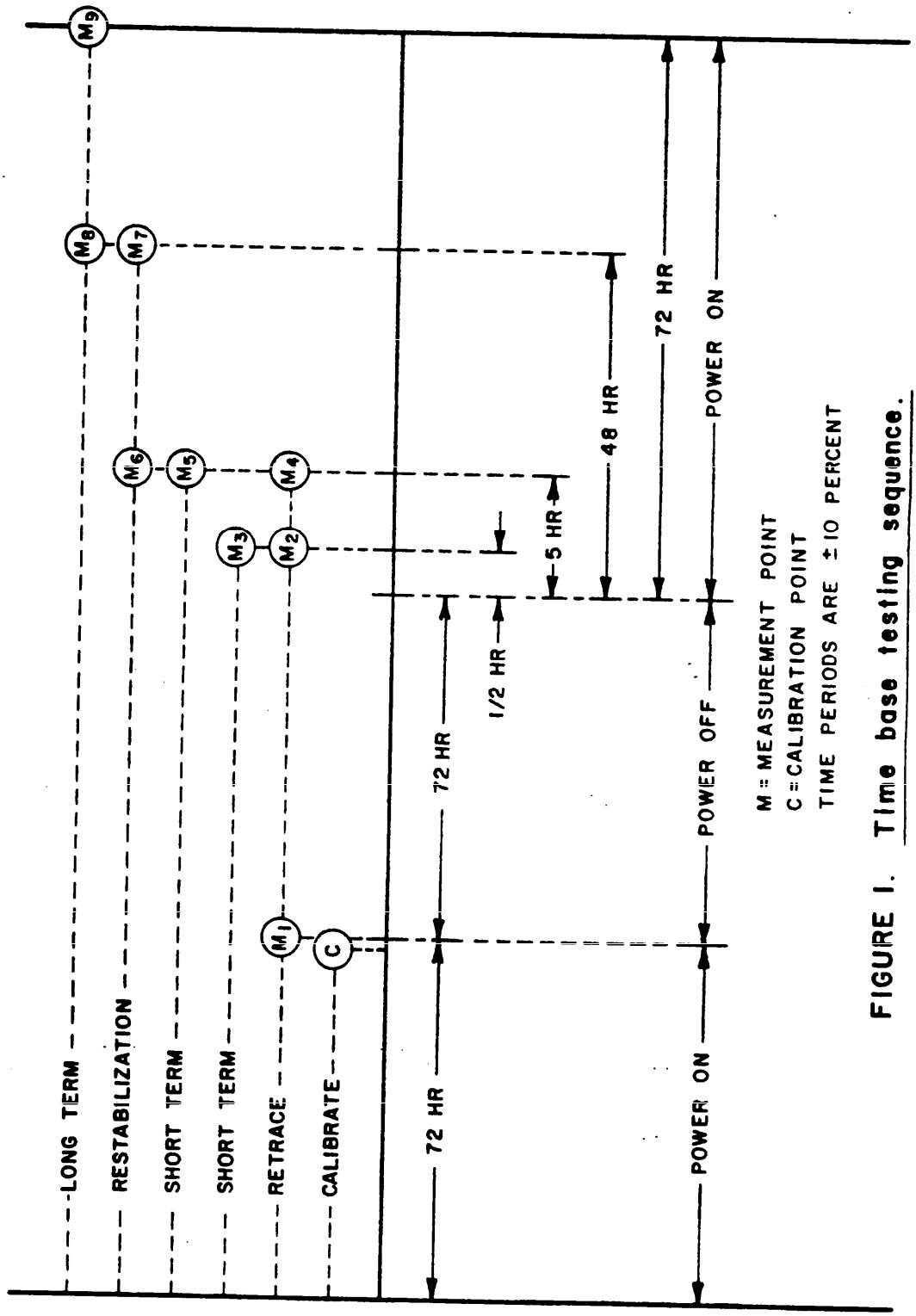


FIGURE 1. Time base testing sequence.

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4.5.2 Alternate power source testing. Testing for the alternate power sources shall be in accordance with the electrical power tests of MIL-T-28800 except the alternate power source shall be used.

4.6 Optional quality assurance testing. For Type III equipment and the Type III portion of Type II equipment, the procuring activity may substitute a Department of Defense (DoD) approved supplier's quality assurance system for any or all tests in Table I.

5. PACKAGING

5.1 Preservation, packaging. Preparation for delivery shall be in accordance with MIL-T-28800 unless otherwise specified by the procuring activity (see 6.2.1.13).

6. NOTES.

6.1 Intended use. Equipment described by this specification are intended for use in testing equipment and systems in the world-wide natural and controlled environments in which military equipment are operated (on a continuous or intermittent basis for extended periods of time), stored or transported, including utilization in all types of moving military and commercial vehicles on land, sea, and in air. The extent of an equipment's penetration (levels of use) into the military environment is limited by its configuration (Type, Class, Style, and Color). Selection of the correct configuration is paramount to ensure procurement of the required equipment. This specification can be used to procure commercial test equipment (Configuration D), or commercial test equipment modified for, or conforming to, specific military requirements (Configurations A, B, and C). The environmental classes and other requirements of this specification represent the majority of clearly discernable levels of the military environment within which an equipment will be used.

6.2 Ordering data. The procurement documents should specify:

6.2.1 Procurement requirements.

6.2.1.1 Title, number, and date of this specification.

6.2.1.2 Configuration desired. (See 1.2 and 3.1).

6.2.1.3 Finishes of Configuration A if other than forest green (see 3.3.3.1).

6.2.1.4 Power cables if operation from 230V is desired. (see 3.5.5.1b and 3.5.3.2b).

6.2.1.5 Type of rackmount slide for Configuration C (see 3.7.3.1).

6.2.1.6 EMI option if other than 3.9.5.1.

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6.2.1.7 Maintenance and calibration aids option (see 3.11.2.1).

6.2.1.8 Time base option if other than 3.12.3.1 (see 3.12.3.2).

6.2.1.9 Logic interface option (see 1.2 and 3.12.13).

6.2.1.10 First Articles, when required (see 4.2.1).

6.2.1.11 Bid samples, when required (see 4.2.2).

6.2.1.12 Levels of packing and packaging (see 5.1).

6.2.1.13 Bid Sample test location (see 4.3.2).

6.2.1.14 Warranty.

6.2.2 Contract data requirements.

6.2.2.1 Nomenclature assignment (see 3.8 and 3.11.2.1).

6.2.2.2 National stock number (see 3.8).

6.2.2.3 Bid Sample test plans, when required (see 4.3).

6.2.2.4 First article test plans, when required (see 4.3).

6.2.2.5 First article test reports, when required (see 4.3).

6.2.2.6 Test equipment approval requests (see 4.5).

6.2.2.7 Pretest performance records (see 4.5).

6.2.2.8 Maintainability test records (see 4.5).

6.2.2.9 Procedure for water-washing equipment (see 3.11.3).

6.2.2.10 Reliability test records (see 4.5).

6.2.2.11 Quality assurance test plans (see 4.4).

6.2.2.12 Quality assurance test reports (see 4.4).

6.2.2.13 Tilt bail handle approval, Configuration B (see 3.7.2).

6.2.2.14 Fungus inert materials certification (see 3.9).

6.2.3 Acquisition management systems.

6.2.3.1 Reliability (see 3.10.1).

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6.2.3.2 Maintainability (see 3.11).

6.2.3.3 Calibration system (see 4.5.).

6.3 Definitions.

6.3.1 Fractional frequency error. Fractional frequency error is defined as the change (produced by any cause) in frequency divided by the average frequency (before the cause). For example in determining the line voltage stability, if the standard output frequency were 1 X 10^7 MHz and if the standard output frequency varied by 10 Hz for any line voltage change within the specified steady state band then the ffe for the line voltage would be 10 divided by 1 X 10^7 which is equal to 1 X 10^{-6}.

6.3.2 Short term stability. The short term stability is defined as the square root of the Allan variance of the oscillator under test divided by the average frequency of the oscillator under test. The band width shall be 30 kHz.

f_o = average frequency of oscillator under test.

$\bar{f}_k$ = k^{th} frequency sample (a frequency sample is a sample of the frequency averaged over 1 second)

M = quantity of adjacent frequency samples. (M = 40 for this specification)

$$\text{Short term stability} = \frac{1}{f_o}\sqrt{\frac{1}{2(M-1)}\sum_{k=1}^{M-1}\left(\bar{f}_{k+1} - \bar{f}_k\right)^2}$$

6.3.3 Sample rate. Sample rate is the time interval between the trailing edge of one gate pulse and the leading edge of the next gate pulse.

6.3.4 Front panel sample. A front panel sample is without holes, is 100mm (4 inches) by 100mm (4 inches), and is marked with the same marking process to be used on production equipment.

6.4 Bid sample rationale.

6.4.1 Reliability rationale. Reliability rationale submitted with the Bid Sample should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar equipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accomplished with specifications and technical literature on the similar equipment. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production reliability testing.

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6.4.2 Maintainability rationale. Maintainability rationale submitted with the Bid Sample should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design features relevant to maintainability may also be submitted (see 3.4). The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production maintainability testing.

Custodians:
Air Force - 99
Army - EL
Navy - EC

Review Activities:
Army - MI
Navy - SH, AS

User Activities:
Army - MU, SC
Navy - OS

Preparing Activity:
Air Force - 99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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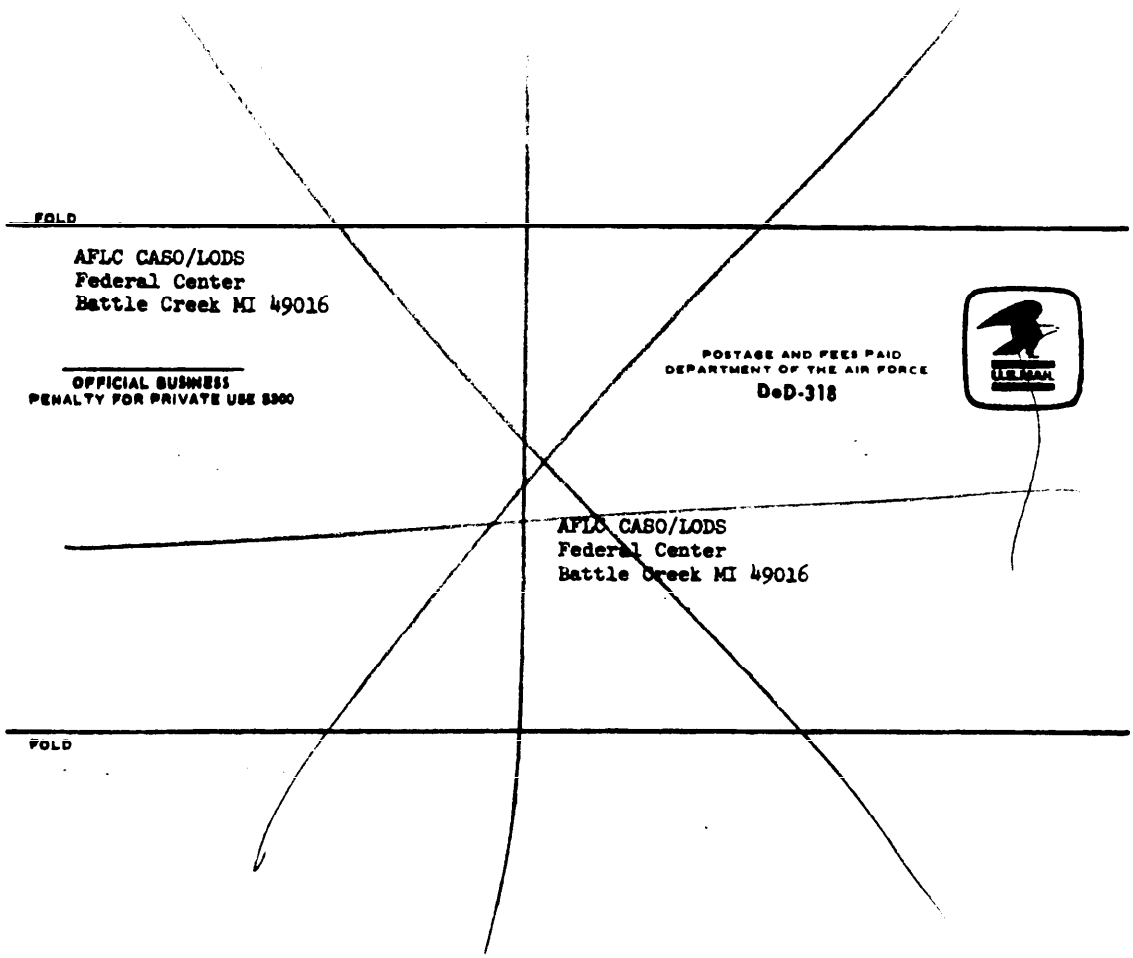
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